

**Fifth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

DATABASE MANAGEMENT SYSTEMS

Time : 3 Hours]

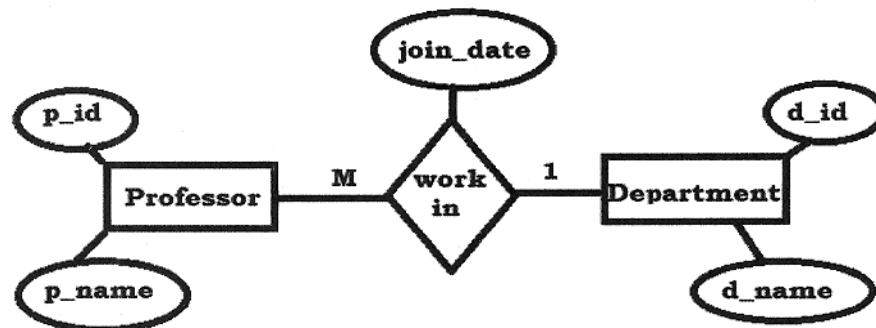
[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) Draw a neat diagram wherever necessary.

1.
 - (a) Explain various architectures of DBMS. 3(CO1)
 - (b) Explain ACID property of DBMS. 4(CO1)
 - (c) Justify the need for Data abstraction in DBMS. Also name various levels of data abstraction. 3(CO1)

2. (a)



3(CO4,5)

- (b) Consider the universal relation $R=\{A,B,C,D,E,F,G,H,I,J\}$ and the set of functional dependencies :

FD: $\{A, B\} \rightarrow \{C\}; \{A\} \rightarrow \{D, E\}, \{B\} \rightarrow \{F\}, \{F\} \rightarrow \{G, H\}, \{D\} \rightarrow \{I, J\}.$

Determine all possible Super keys, candidate keys and a primary key.

7(CO4)

3. Design a database considering the following database schema–
 Student (Roll_No, S_name, Address)
 Teacher (T_ID, T_name, T_course, Roll_No)
 College (C_ID, C_address, Roll_No)
 Write SQL queries or Relational algebra expression for the following :
- (a) Insert a tuple into relation Student.
 - (b) Find the names of all the students who live in Nagpur ?
 - (c) Find the name of the teacher who teaches DBMS course ?
 - (d) Find the name of the teacher who teaches DSP course to student named 'XXXX' ?
 - (e) Find the name of the students who took the course taught by the teacher named 'YYYY' ?
 - (f) Find the name of the student who enrolled in the college having ID 12345.
 - (g) Delete record of students whose address is Mumbai. 10(CO3,5)
4. (a) Explain the need for indexing. Also explain various types of indexing. 5(CO4)
- (b) Write a short note on Query Optimization. 5(CO2)
5. (a) Consider a hash table of size seven, with starting index zero, and a hash function, $h(K) = (3K + 4)$. Assuming the hash table is initially empty, determine the contents of the table when the sequence 1, 3, 8, 10 is inserted into the table using closed hashing ? 5(CO2)
- (b) What is query processing ? Explain the steps involved in query processing and explain it in detail. 5(CO2)
6. (a) Explain Transaction and Schedule with suitable example. 5(CO1)
- (b) What is the two-phase locking protocol ? Two-phase locking protocol guarantees Serializability. Justify. 5(CO1)

